

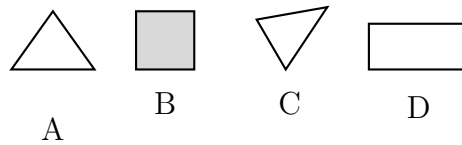
Sort Shapes Like a Mathematician

Sorting shapes by sides and corners, not by color or size

Directions:

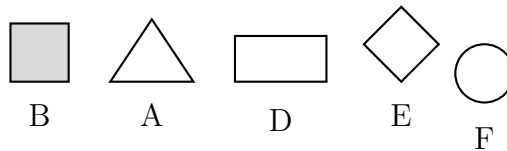
- Sort by things you can COUNT: sides and corners.
- Color, size, and tilt never decide a shape's group.

1. Watch how Ada sorts: she ignores color and tilt, counts the straight sides of each shape, and groups the equal counts. Now you try with these shapes:



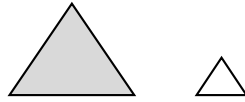
- (a) How many shapes have exactly 3 sides? _____
(b) Write their letters: _____

2. Look at these five shapes:



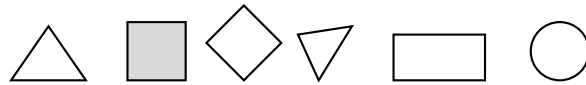
- (a) How many shapes have exactly 4 corners? _____
(b) Write their letters: _____

- 3.** Kim looks at a big gray triangle and a small white triangle. She says, “They cannot go in the same group — one is big and one is small!”



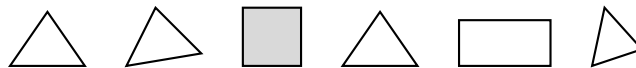
- (a) How many straight sides does EACH shape have? _____
(b) Is Kim right? Say why or why not.

- 4.** Sort these six shapes by their number of sides:



- (a) 3-sided shapes: _____ (b) 4-sided shapes: _____
(c) How many shapes fit in NEITHER group? _____

- 5.** Sort these six shapes by corners, then compare the groups:



- (a) Shapes with 3 corners: _____ (b) Shapes with 4 corners: _____
(c) Write $>$ or $<$: your (a) count _____ your (b) count

- 6.** Raj sorted shapes by color: 3 shapes in his gray group and 5 shapes in his white group. The teacher says color is not a DEFINING attribute.
- (a) How many shapes did Raj sort in all? _____
- (b) Say what Raj should count instead of color when he sorts.

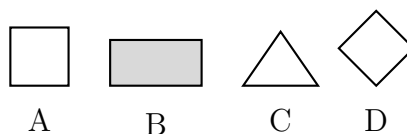
- 7.** Count the groups in this set:



The 5-sided shapes are pentagons.

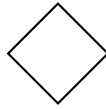
- (a) 4-sided shapes: _____ (b) 5-sided shapes: _____
- (c) How many more 4-sided shapes than 5-sided shapes? _____

- 8.** One of these shapes is the odd one out:



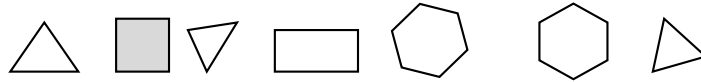
- (a) The odd one out has _____ corners.
- (b) Write its letter: _____

9. The group rule is: “exactly 4 equal sides.” Lily says the tilted shape below is in the wrong group because it looks like a diamond.



- (a) How many equal sides does the tilted shape have? _____
(b) Does it belong in the group? Say why or why not.

10. Challenge! Sort ALL seven shapes by sides, then check yourself:



- (a) 3-sided: _____ (b) 4-sided: _____ (c) 6-sided: _____
(d) Self-check: add your three counts. Do they make all 7 shapes? Sum: _____